

Architecture Proof Boundary

CURRENT STATE AND FUTURE STRATEGY

- Hosted / and /admin are live
- Public proof: same-day 20x20 / 72h full-grid cell batch from 2026-05-08
- Hosted authenticated admin smoke succeeded on May 8, 2026
- Metadata and preview assets use Avalanche Insight Hub branding
- production_scoring_allowed=false and himalayan_accuracy_claim_allowed=false remain the governing boundaries for new evidence lanes

Evidence level: Hosted production plus Repo/admin verified

INTERPRETATION

CURRENT STATE

Evidence now

Use only hosted, admin, repo, and artifact proof that exists today.

GATED

Promotion rules

MTS-LSTM, SAR, TreeSHAP refresh, and runout physics require explicit evidence before promotion.

FUTURE STRATEGY

Validation program

Scientist review decides which technical paths become stronger operational claims.

Current Platform Layers

CURRENT STATE AND FUTURE STRATEGY

- Interface: React, Vite, Leaflet, Recharts
- Data and storage: Supabase, Postgres, Storage, Auth
- Artifact delivery: manifests, hourly grids, runout overlays
- Batch compute: Python jobs
- Governance: model status, benchmark, release gates
- Research evidence: Swiss reproduction artifacts and Himalayan partner-package triage outputs

Evidence level: Repo/admin verified

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Interface Layer

CURRENT STATE AND FUTURE STRATEGY

- / public forecast workspace
- /admin operator observability lane
- Stateful route rendering
- Map, bulletin, timeline, share, export, report controls

Evidence level: Repo/admin verified plus Hosted production

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Supabase Data And Storage Layer

CURRENT STATE AND FUTURE STRATEGY

- Postgres structured records
- Supabase authentication
- Supabase Storage artifact references
- Edge Functions for selected server-side tasks
- Row-Level Security for access boundaries

Evidence level: Repo/admin verified

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Forecast Publication And Artifact Hydration

CURRENT STATE AND FUTURE STRATEGY

- Forecast run metadata
- Artifact manifest
- Hourly grid payloads
- Optional runout overlay payloads
- Browser-side loading and decompression

Evidence level: Hosted production plus Repo/admin verified

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Offline Batch Compute Split

CURRENT STATE AND FUTURE STRATEGY

- Current: scheduled/operator-triggered Python jobs
- Future direction: GitHub Actions or lightweight VPS
- Heavy work: feature selection, rare-event balancing, inference, runout physics
- Public route: presentation and review

Evidence level: Repo/admin verified plus Artifact/doc proof only

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Active Scorer And Explanation Layer

CURRENT STATE AND FUTURE STRATEGY

- Active: surrogate_rf_v1
- Current run explanation: heuristic_fallback
- TreeSHAP: implemented path, not stronger proof for the current active run
- Supporting training concepts: KMeansSMOTE, Recursive Feature Elimination, calibration
- Candidate: MTS-LSTM, gated

Evidence level: Repo/admin verified

INTERPRETATION

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FUTURE STRATEGY

Validation program

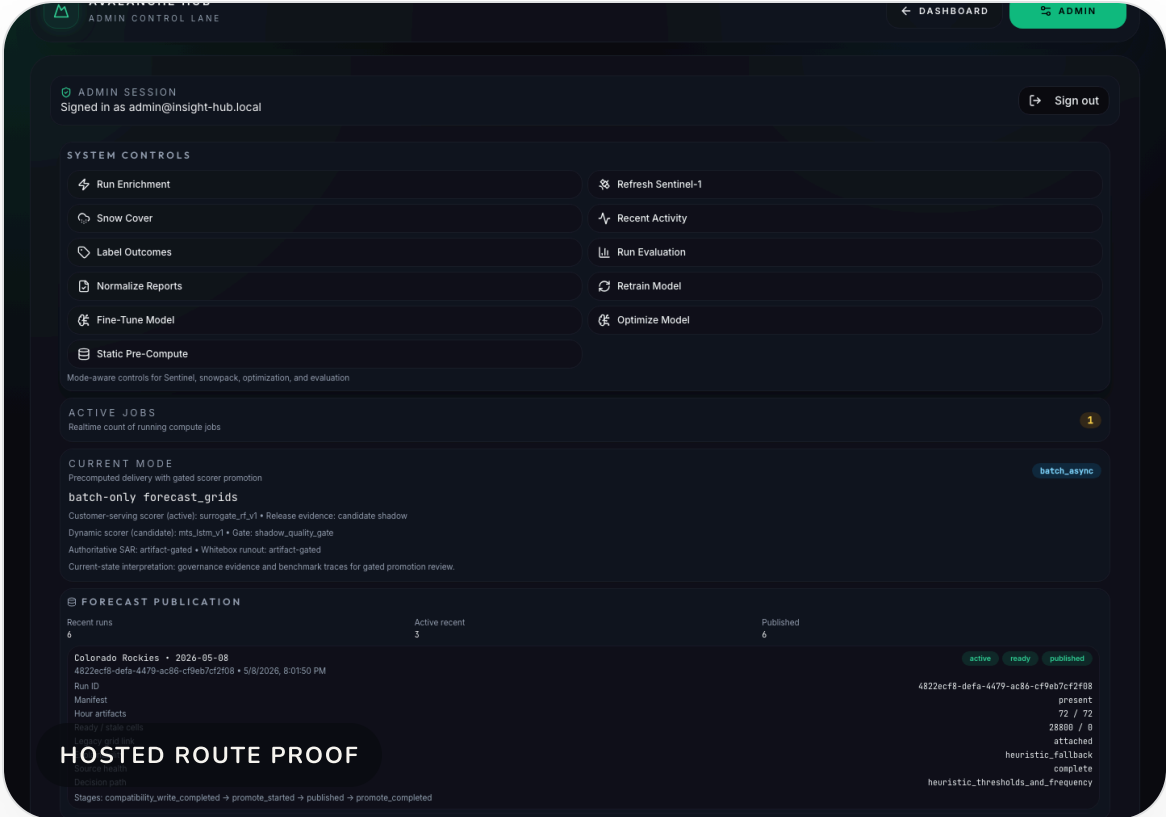
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Admin Governance Lane

CURRENT STATE AND FUTURE STRATEGY

- Source health
- Decision provenance
- Model status
- Stability and benchmark summaries
- Publication controls

Evidence level: Hosted production plus Repo/admin verified



Offline Field Reporting

CURRENT STATE AND FUTURE STRATEGY

- Progressive Web App pattern
- Service Worker registration
- IndexedDB local queue
- Startup and reconnect flush
- Field proof remains deployment-specific

Evidence level: Repo/admin verified

INTERPRETATION

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Modal.com

Candidate

Compute

Plane

CURRENT STATE AND FUTURE STRATEGY

- MTS-LSTM training candidate
- SAR segmentation candidate
- Persistent artifact volumes
- Authenticated worker endpoints
- Current public scorer remains Random Forest baseline
- No GPU run is authorized by partner handoff or synthetic validation success

Evidence level: Repo/admin verified

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Future Offline-Batch Paradigm

CURRENT STATE AND FUTURE STRATEGY

- Presentation: React, Leaflet, 3D views
- Storage: Supabase and Postgres
- Compute: GitHub Actions or lightweight VPS
- Heavy math: feature selection, KMeansSMOTE, inference, runout physics

Evidence level: Artifact/doc proof only

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Proposed Data Contract

CURRENT STATE AND FUTURE STRATEGY

- Public path: avalanche_events, topographic enrichment, forecast_grids, grid_geojson, and runout_polygons
- Himalayan v3 path: ten evidence CSV templates, partner_source_manifest, source_ref hashes, and release-gate attestations
- Partner packet: partner_handoff_readme.md, partner_field_dictionary.md, checksum guide, source manifest template, and blank evidence CSVs
- Triage outputs: source-manifest validation, evidence validation, leakage audit, artifact manifest, source traceability, and status dashboard
- PostGIS remains a proposed geospatial extension, not current partner-evidence proof

Evidence level: Artifact/doc proof only

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Proposed ML And Runout Pipeline

CURRENT STATE AND FUTURE STRATEGY

- KMeansSMOTE for rare events
- Recursive Feature Elimination for feature discipline
- Calibrated Random Forest
- Swiss RAvaFcast research lane: Stage 1 RF4, Stage 2 GPxyz blocked until station coordinates, Stage 3 aggregation blocked until grid/polygons
- Tree variance for uncertainty bounds
- WhiteboxTools Alpha-Beta runout with DEM cropping remains a validation path
- Current active runout proof is analytical Alpha-Beta fallback; Whitebox smoke passed separately

Evidence level: Artifact/doc proof only

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Current, Partial, Candidate, Proposed

CURRENT STATE AND FUTURE STRATEGY

- Current: hosted public route, admin route, RF baseline, artifact hydration
- Partial/current: offline report queueing
- Candidate/gated: MTS-LSTM, SAR, TreeSHAP refresh for the active full-grid run
- Research-only: Swiss RAVaFcast reproduction and Himalayan partner-package triage
- Proposed/validation: GitHub Actions/VPS batch compute, PostGIS terrain enrichment, WhiteboxTools runout

Evidence level: Repo/admin verified plus Artifact/doc proof only

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Architecture Gates And Next Workstreams

CURRENT STATE AND FUTURE STRATEGY

- Full-grid same-day publication with structured bulletin content is current
- Benchmark packaging
- Admin proof refresh before each customer-send reuse
- Generate and send the v3 handoff packet without synthetic rows
- Run `run_himalayan_partner_package_triage` only after real SASE/DGRE evidence arrives
- Schema and PostGIS migration decision
- Offline-batch execution proof
- TreeSHAP, SAR, and Whitebox runout validation gates

Evidence level: Artifact/doc proof only

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